

Indexing

An **index** in a database is like the **index at the end of a textbook**.

- **Without an index:** to find "Chapter 7", you would flip through **every page** until you see it (slow).
- **With an index:** you look in the index, it tells you the **exact page number**, and you jump there (fast).

In database terms

- A **table** can have millions of rows.
- If you run a query like:

```
WHERE rfq_id = '...'
```

the database needs to find matching rows.

No index → it checks row 1, row 2, row 3... (called a *full scan*).

With index on rfq_id → it jumps مباشرة to the matching rows (much faster).

Why indexes exist

To make frequent queries fast, especially:

- filtering (`WHERE`)
- sorting (`ORDER BY`)
- joining tables (`JOIN`)

The trade-off (important)

Indexes make **reads faster**, but they make **writes a bit slower** (because DB must update the index when you insert/update/delete).

That's the whole concept.